

AI Token Global

Final Project Audit Report

AI Token Global — Final Project Audit Report

Date: 2026-05-08 **Prepared by:** Project Supervisor (synthesizing technical, SEO, automation, and QA/mobile audits — four specialists, first inclusion of QA/mobile) **Audience:** Project sponsor / decision-maker **Decision asked:** Continue resource commitment to AI Token Global? **Supersedes:** [audits/FINAL_PROJECT_AUDIT_2026-05-07.md](#) (archived for diff comparison)

0. Project Context (for readers new to the project)

This section orients a reader who has not worked on the project before. Returning readers can skip ahead to Section 1.

What the site is

AI Token Global is a multilingual content hub being built to teach a Western, English-speaking audience about AI tokens, model pricing, API choices, use cases, and best practices. It is the international expansion of an existing Taiwanese brand — **AI Token King** — whose Traditional Chinese site lives at `aitoken.com.tw`. The Chinese site is the content source: copy is translated and adapted (not invented) for a Western audience, with a redesigned look. The long-term plan is to support 10–15 languages, with English and Spanish as the launch pair.

The brand mascot is a corgi with a crown, the primary brand color is purple `#6155F1`, and the design system pairs Kanit (headings) with Plus Jakarta Sans (body). A strict "anti-generic" set of rules in `CLAUDE.md` — no default Tailwind blue/indigo, no `transition-all`, layered shadows and gradients only — gives the site its visual identity.

Tech stack (locked in)

The architecture decision was made deliberately and is not open for revision. Any audit recommendation argues *inside* these constraints:

- **Astro** as the static-site framework, generating pages at build time
- **Sanity CMS** as the content store (project ID `mq3wxr8n`, dataset `production`)
- **AWS Amplify** for hosting (CloudFront-backed CDN)

The repo is `antonioduran-insight/AI_Token_Global` on GitHub, with the Astro project at the root and a Sanity Studio inside `studio/`. The validated content pattern: one Sanity document per (page × language), Portable Text for body fields, and `[lang]` dynamic routes that loop over a `SUPPORTED_LANGS` constant — meaning a new language should require *one constant change plus content entry*, with no per-page code edits.

How content flows

Editors work in Sanity Studio (`localhost:3333` for local dev). They publish documents tagged with a `language` field. Astro fetches Sanity at build time and statically generates every language variant. So `/en/ai-trends` and `/es/ai-trends` are different documents that share one `.astro` template. **A current gap:** no webhook from Sanity to Amplify exists yet, so a content publish does not trigger a rebuild — the live site stays stale until the next `git push`. Wiring this webhook is the single biggest pending leverage point in the project.

Where things stand (May 2026, after 12 work sessions)

Done: repo restructure (Astro at root, HTML prototypes archived); dynamic language switcher in Nav and Footer; Sanity schema POC built and validated on the AI Trends page (the pattern everything else will follow); blog routes working with three English posts live; AWS Amplify deployment guide written.

Not done: 10 of 14 planned pages still exist only as HTML prototypes in `archive/` — they need to be ported to Astro using the Sanity singleton pattern (this is the next active task); homepage is a "Coming Soon" placeholder; the site has not been deployed to Amplify yet; no SEO infrastructure exists (no `robots.txt`, no sitemap, no `hreflang`, no Open Graph tags, no SEO fields in any Sanity schema); languages 3–15 not yet planned in detail.

The 14 pages

Four are working: `/en/`, `/es/` (placeholders), AI Trends, blog index, blog post template. Ten remain to port, batched as **Batch A** (four API model pages — `api-compare`, `chatgpt-api`, `claude-api`, `gemini-api`), **Batch B** (five guide pages — `beginners-guide`, `user-guide`, `use-cases`, `token-calculator`, `compliance`), and **Batch C** (the homepage, saved for last because it is most complex).

Team and budget shape

The team is small — implied as roughly one technical operator plus one editor. Cost trajectory is friendly: ~\$1.25/month today (just the domain), projecting to \$26–34/month at 12 months (5+ languages, ~50K visitors), and capping around \$50–60/month at full 15-language scale. The largest hidden cost is *manual* content entry: without bulk-migration scripts, importing the 200+ existing articles would take 80–120 hours; with scripts, ~10–13 hours.

Constraints anyone joining should know up front

- **The architecture is not open for debate.** Astro + Sanity + Amplify, `[lang]` routing, Sanity-first content, and Portable Text are decided. Implementation gaps inside those choices are fair game; the choices themselves are not.
- **Sanity-first means no hardcoded English in `.astro` files.** All UI strings live in `src/i18n/{en,es}.json` or in Sanity. The audit found several violations where Spanish users still see English UI labels.
- **Anything assuming exactly EN+ES will silently break at language 3.** The audit flagged two specific places (a `locale` ternary in the blog pages and a static `language` list in the post schema) that must be fixed before scaling.
- **Mobile traffic is expected to dominate.** The QA/mobile audit added in this synthesis (2026-05-08) found that mobile navigation is currently completely broken on phones and tablets — a launch blocker the previous audit cycle did not catch. Centralizing the responsive system in `global.css` before Task #5 stamps the singleton pattern across 11 pages is now a pre-flight item.

1. Executive Summary

Recommendation: GO WITH CAVEATS — unchanged from the May 7 verdict. The QA/mobile audit added in this synthesis **reinforces and extends** the May 7 conclusion rather than shifting it. The architecture is still sound; the team has shipped real progress; the new findings (one of which is a launch blocker the previous cycle missed) all fit inside the existing pre-flight + Task #5 framework. Crucially, the QA audit caught these issues *before* Task #5 stamps the singleton pattern across 11 pages — the exact value the audit cycle is meant to produce. The 30-day priority list expands by roughly half a day; everything else holds.

What changed since May 7

- A **fourth specialist audit (QA/mobile)** was run on 2026-05-08 and is included in this synthesis for the first time.
- Risk Matrix gains a new category: **Visual / Mobile / A11y**.
- One **new Bucket A (launch-blocking) cluster** emerged: mobile navigation is completely broken on every phone and tablet — a CSS `!important` rule overrides the JavaScript that tries to open the menu (Q-01/Q-02). The FAQ accordion on the AI Trends page is also broken in production today (Q-03).
- One **new pre-flight item (Task #0f)** is recommended for the implementation plan: a centralized responsive breakpoint system in `src/styles/global.css` plus the mobile/A11y fixes the audit surfaced. ~4–6 hours of work; absorbs eleven follow-on copies of the same problem if done before Task #5.
- The May 7 cost trajectory and architecture verdict are unchanged.

Top 3 Strengths (unchanged)

1. **Core architecture is correctly implemented and locked in.** Dynamic `[lang]` routing, Sanity-first content, `getStaticPaths()` over `SUPPORTED_LANGS`, and the language switcher all work as designed. There are no build-breaking issues. The Astro + Sanity + AWS Amplify stack remains the right choice for a 15-language content hub at this budget.
2. **Sanity schema quality is high and the POC pattern is reusable.** Both `post` and `aiTrendsPage` use Portable Text for body fields, require `language`, and reuse `faqItem` correctly. With the pre-flight expansions (SEO, EN/ES delocalization, responsive system) the pattern can scale to the remaining 11 page schemas without redesign.
3. **Cost trajectory is highly favorable.** Current burn is roughly \$1.25/month (domain only). Projected burn at 12-month scale is \$26–34/month; full 15-language scale stays under \$60/month.

Top 4 Risks (was 3 — Visual/Mobile/A11y added)

1. `.env` is committed to git with the live Sanity project ID. Security and quota-abuse exposure that contradicts the file's own warning header. **Must fix before the next push.** (*Unchanged from May 7.*)
2. **Zero SEO infrastructure exists today** — no `robots.txt`, no sitemap, no canonical, no `hreflang`, no Open Graph, no `seo` fields in any Sanity schema. The `seo` object must land in the schema pattern *before* Task #5 replicates it. (*Unchanged from May 7.*)
3. **Two hidden EN/ES assumptions will silently break at language 3.** The `locale` ternary in both blog pages and the static language list inside `studio/schemas/post.ts`. Both block Task #10. (*Unchanged from May 7.*)
4. **Mobile navigation is broken on every phone and tablet, the FAQ accordion does not open, and the entire CSS system is desktop-first despite `CLAUDE.md` declaring "mobile-first."** **NEW.** A CSS `!important` rule in `global.css:277` (`.desktop-nav { display: none !important }`) overrides the inline `style.display = 'flex'` toggle in `Nav.astro:118`, so the hamburger button has no visible effect. The FAQ accordion's `max-height` transition is never triggered because the JavaScript adds `.open` to the wrong element. Two CSS animations (`transition: width` on the blog reading bar; `transition: max-height` on the FAQ) violate `CLAUDE.md`'s `transform / opacity`-only rule. The audit also caught one WCAG AA color contrast failure in the footer (`#666` on `#1C1C1C` $\approx 3.0:1$; needs $\geq 4.5:1$).

30-Day Priorities (expanded)

- **Week 1: Pre-flight (Task #0).** Originally 5 sub-items; now 6 with the QA additions. Remove `.env` from git (#0a), add the `seo` object to schemas (#0b), wire `BaseLayout.astro` for canonical/hreflang/OG (#0c), install sitemap + robots (#0d), fix the EN/ES delocalization (#0e), **and now: standardize the responsive breakpoint system + fix the mobile nav, FAQ accordion, illegal transitions, missing `:active` states, hamburger touch target, and footer contrast (#0f, NEW).** Doing this *now* — before Task #5 replicates the singleton pattern across 11 pages — avoids fixing the same problem 11 times. Total Task #0 effort: ~1.5–2.5 days (was 1–2).
- **Week 2: Deploy infrastructure (Task #8).** Add `amplify.yml`, set Amplify Console env vars, configure the Sanity → Amplify webhook. Hard-blocked on Task #0a until `.env` is untracked.
- **Week 3: Bulk migration scripts (Task #6).** `scripts/upload-images.js` + `scripts/convert-articles.js` — converts ~80–120 hours of manual entry into ~10–13 hours.
- **Week 4: Begin Task #5 page port.** Start with Batch A (the four API model pages) using the SEO-hardened **and** mobile-hardened singleton pattern from Week 1.

Cost Trajectory (unchanged)

Stage	Languages	Traffic	Build Activity	Monthly Cost
Today (pre-launch)	2 (EN + ES)	0	0 builds	~\$1.25
Month 6	2	~5K visitors	~10 publishes/day	~\$3–5
Month 12	5+	~50K visitors	~20 publishes/day	~\$26–34
Full scale	15	Daily publishing	All langs active	~\$50–60

2. Risk Matrix

All findings from the four specialist audits, grouped by impact tier and now by category. **A new "Visual / Mobile / A11y" category** is added to accommodate the QA audit. Severity is the supervisor's judgment after cross-checking.

Bucket A — Blocking Launch (cannot deploy or will visibly break)

ID	Finding	Category	Source	Effort	Owner Task
A-01	<code>amplify.yml</code> does not exist at repo root; build spec not version-controlled	Automation	Automation	S (1 hr)	#8
A-02	Sanity → Amplify publish webhook entirely unconfigured — content edits will not trigger rebuilds	Automation	Automation	S (30 min)	#8
A-03	Amplify Console env vars not set — first deploy will return empty Sanity queries with no error	Automation	Automation	S (10 min)	#8
F-01	<code>.env</code> committed to git with live <code>mq3wxr8n</code> project ID despite gitignore policy	Technical	Technical	S (5 min)	#0a (security gate)
S-01	No <code>public/robots.txt</code> — search engines have no crawl instructions or sitemap pointer	SEO	SEO	S (15 min)	#0d
S-04	<code>astro.config.mjs</code> has no <code>site</code> URL and no sitemap integration installed	SEO	SEO	S (30 min)	#0d

ID	Finding	Category	Source	Effort	Owner Task
S-02 / S-03 / S-05	No <code>hreflang</code> tags anywhere; sitemap won't include hreflang clusters	SEO	SEO	M (3–4 hr)	#0c
S-06	No <code><link rel="canonical"></code> on any page	SEO	SEO	S (30 min)	#0c
S-07 / S-08	No Open Graph or Twitter Card tags — every social share produces a blank preview	SEO	SEO	M (2–3 hr)	#0c
S-18	Coming Soon homepage will be indexed with thin content; no <code>noindex</code> mechanism	SEO	SEO	S (30 min)	#0b
Q-01	<code>!important</code> on <code>.desktop-nav { display: none }</code> overrides the inline-style toggle — mobile menu cannot open	Visual / Mobile / A11y	QA/Mobile	S (15 min)	#0f (NEW)
Q-02	No mobile nav drawer styles exist — even if Q-01 is fixed, the panel has no mobile layout	Visual / Mobile / A11y	QA/Mobile	M (1–2 hr)	#0f (NEW)
Q-03	AI Trends FAQ accordion never opens — JS adds <code>.open</code> but CSS <code>max-height</code> transition is on the wrong selector	Visual / Mobile / A11y	QA/Mobile	S (30 min)	#0f (NEW)
Q-04	Possible horizontal scroll on mobile — needs verification once nav is repaired	Visual / Mobile / A11y	QA/Mobile	S (30 min)	#0f (NEW)
Q-12	Footer secondary text fails WCAG AA contrast (<code>#666</code> on <code>#1C1C1C</code> $\approx 3.0:1$; needs $\geq 4.5:1$)	Visual / Mobile / A11y	QA/Mobile	S (5 min)	#0f (NEW)

Total launch-blocker count: 14 (was 9). The five new entries are the QA/mobile cluster. Of these, Q-01–Q-03 are functional bugs that will be visible to any phone user on day one of launch. Q-12 is a legal/accessibility risk (ADA-style claims have been filed against contrast failures). All five fit inside Task #0f and are bounded.

Bucket B — Blocking Scale (works at EN+ES, breaks at language 3+ or 200+ articles)

ID	Finding	Category	Source	Effort	Owner Task
F-05	<code>locale</code> ternary defaults to <code>en-US</code> when language 3 is added	Technical	Technical	S (15 min)	#0e (#10 prereq)
A-07	<code>studio/schemas/post.ts</code> <code>language</code> field is a static <code>['en', 'es']</code> list	Automation	Automation	M (2 hr or install plugin)	#0e (#10 prereq)
S-09	No <code>seo</code> object in any Sanity schema — must be added to base pattern <i>before</i> it is replicated to 11 pages in Task #5	SEO	SEO	M (1 hr per schema)	#0b
F-07	<code>ai-trends.astro</code> renders blank silently if Sanity returns <code>null</code>	Technical	Technical	S (10 min)	#0f / #5 (pattern)
S-14	Language switcher always routes to <code>/lang/</code> instead of equivalent page	SEO	SEO	M (3 hr)	#10
A-05 / A-06	No bulk migration scripts — 200+ articles will need 80–120 hours of manual Sanity entry	Automation	Automation	M (10–13 hr to build, saves ~110 hr)	#6 (scope expansion)
A-09 / A-10	No AI-assisted meta-description or alt-text generation	Automation	Automation	M (1 day each)	#6 / #12
Q-05	Hamburger button hit area ≈38px — below 44×44px touch-target minimum (Apple HIG / WCAG)	Visual / Mobile / A11y	QA/Mobile	S (5 min)	#0f (NEW)
Q-07	Two illegal CSS transitions violate <code>CLAUDE.md</code>'s <code>transform / opacity</code> rule: <code>transition: width</code> on blog reading bar; <code>transition: max-height</code> on FAQ	Visual / Mobile / A11y	QA/Mobile	S (30 min)	#0f (NEW)

ID	Finding	Category	Source	Effort	Owner Task
Q-08	All CSS is desktop-first (<code>max-width</code> queries) despite <code>CLAUDE.md</code> declaring "mobile-first" — entire system is override-down, not enhance-up	Visual / Mobile / A11y	QA/Mobile	M (3–4 hr)	#0f (NEW — centralization)
Q-09	Missing <code>:active</code> states on <code>.dropdown-item</code> and <code>.faq-question</code> — <code>CLAUDE.md</code> requires hover, focus-visible, and active on every interactive element	Visual / Mobile / A11y	QA/Mobile	S (15 min)	#0f (NEW)
Q-13	Blog images use raw <code></code> with no <code>width</code> / <code>height</code> or <code>aspect-ratio</code> — CLS risk that compounds when Task #5 replicates the blog pattern	Visual / Mobile / A11y	QA/Mobile + SEO (S-17 cross-ref)	M (1 hr)	#0f / #9 (post-deploy migration to Astro <code><Image></code>)
Q-17	Conflicting breakpoints — Footer uses 768px scoped <code>!important</code> while <code>global.css</code> uses 640px — single mobile breakpoint must be centralized	Visual / Mobile / A11y	QA/Mobile	S (15 min)	#0f (NEW)

Bucket C — Quality and Optimization (improves but does not block)

ID	Finding	Category	Source	Effort	Owner Task
F-02 / F-03 / F-04	Hardcoded English strings in <code>[lang]/index.astro</code> , <code>ai-trends.astro</code> , <code>blog/[slug].astro</code>	Technical	Technical	S (1 hr total)	#0 / #5
F-06 / S-16	<code>coverImage</code> has no alt-text field; inline image alt has no <code>Rule.required()</code>	Technical / SEO	Technical / SEO	S (15 min)	#0 / #5
F-08	Hard-coded <code>https://aitokenglobal.com</code> in blog slug page should use <code>Astro.site</code>	Technical	Technical	S (10 min)	#9

ID	Finding	Category	Source	Effort	Owner Task
F-13	Default <code>description</code> prop in <code>BaseLayout.astro</code> is English-only	Technical	Technical	S (15 min)	#0 / #5
S-13 / Q-14	Google Fonts loaded via render-blocking CSS <code>@import</code> ; no preconnect (cross-confirmed by both audits)	SEO	SEO + QA/ Mobile	S (15 min)	#9
S-17	Raw <code></code> tags everywhere instead of Astro <code><Image></code> (cross-confirms Q-13 from a perf/SEO angle)	SEO	SEO	L (1 day)	#9 / post-launch
A-08	No webhook debounce — every Sanity publish triggers a build	Automation	Automation	S (2 hr)	#8
A-11	No Amplify build-failure notifications	Automation	Automation	S (15 min)	#8
A-12	No AWS Budget alarms for cost spikes	Automation	Automation	S (15 min)	#8
A-13	No Sanity dataset backup automation	Automation	Automation	M (2 hr)	#11 (NEW from May 7)
Q-11	Heading scale on small viewports needs verification — Kanit at large desktop sizes may be too tight on mobile	Visual / Mobile / A11y	QA/Mobile	S (1 hr)	#0f / #9
Q-15	Tailwind CDN script lacks <code>defer</code> ; CDN Tailwind itself is not production-appropriate	Visual / Mobile / A11y	QA/Mobile	S (5 min for <code>defer</code>); M (2 hr for build-step migration)	#0f for <code>defer</code> ; #9 for full migration
Q-16	No <code>prefers-reduced-motion</code> block — animations cannot be disabled by users with vestibular sensitivities	Visual / Mobile / A11y	QA/Mobile	S (15 min)	#0f / #9

Bucket D — Nice-to-Have (defer to post-launch)

ID	Finding	Category	Source	Effort
F-09	Privacy / Terms footer links are dead <code>href="#"</code>	Technical	Technical	S (1 hr)
F-10	Sources download CTA links to <code>href="#"</code> with no file	Technical	Technical	S (15 min)
F-11	Copyright year is 2025; brand name reads "AI Token King" instead of "AI Token Global"	Technical	Technical	S (5 min)
S-10	No <code>Article</code> JSON-LD on blog post pages	SEO	SEO	M
S-11	No <code>FAQPage</code> JSON-LD on AI Trends page	SEO	SEO	M
S-12	No <code>Organization</code> JSON-LD anywhere	SEO	SEO	S
S-15	No <code>BreadcrumbList</code> JSON-LD	SEO	SEO	M
A-14	No <code>node_modules</code> cache key in <code>amplify.yml</code>	Automation	Automation	S
A-15	No branch auto-deploys for preview URLs	Automation	Automation	S
Q-20	Input <code>font-size</code> $\geq 16\text{px}$ to prevent iOS auto-zoom — N/A until newsletter / contact forms exist	Visual / Mobile / A11y	QA/Mobile	N/A

QA Pass Items (no action needed)

The QA/mobile audit confirmed several rules already correctly enforced. Worth recording so the team gets credit:

- **Q-06** — `transition-all` is nowhere in the codebase (CLAUDE.md compliance).
- **Q-10** — **Body font-size is 16px** (no iOS auto-zoom).
- **Q-18** — `aria-label="Toggle mobile menu"` is on the hamburger button (correct ARIA).
- **Q-19** — **No bare** `outline: none` anywhere — focus indicators are preserved.

3. Specialist Audit Highlights

3.1 Technical Audit — Headline (unchanged from May 7)

The core architecture is sound. `[lang]` routing, `getStaticPaths()`, Sanity-first content, and dynamic language switching are correctly implemented. The blockers are concentrated in two places: `.env` is committed to git (a security exposure), and several hardcoded English strings violate the Sanity-first principle and appear on the Spanish site today.

Key findings remain F-01 (`.env` in git), F-05 (`locale` ternary regression risk), F-07 (silent null-render in `ai-trends.astro`), and well-designed Sanity schemas with required `language` validation.

3.2 SEO Audit — Headline (unchanged from May 7)

The site has no SEO infrastructure at all. Launching today would mean search engines crawl blind, English and Spanish pages compete for the same rankings, every social share produces a blank preview, and zero structured data unlocks rich results.

Multilingual readiness scorecard remains 1 of 10 passing (`<html lang>` only). Pre-launch must-fix order: robots.txt → sitemap + `site` URL → canonical → hreflang → OG + Twitter Card → `seo` object in schemas → `noindex` on Coming Soon homepage. With Task #0c/d landing in pre-flight, all eight blockers will be cleared before deploy.

3.3 Automation Audit — Headline (unchanged from May 7)

The architectural automation works. The operational glue is missing: no `amplify.yml`, no Sanity webhook, no env vars in Amplify Console, and no bulk migration scripts. The single biggest leverage point is the Sanity → Amplify webhook — a 30-minute task whose absence means every content publish leaves the live site stale.

Cost projection unchanged. AI-assisted ops roadmap (translation drafting, alt-text generation, meta description, FAQ extraction, content QA) sits in proposed Task #12.

3.4 QA & Mobile Audit — Headline (NEW — first inclusion)

Mobile navigation is completely broken on every phone and tablet today, the FAQ accordion on the AI Trends page doesn't open, and the entire responsive system is desktop-first despite `CLAUDE.md` declaring otherwise. None of the findings are architecturally hard — most are 5–30-minute fixes — but they need to land **before** Task #5 stamps the singleton pattern across 11 pages, because the same broken pattern would otherwise be fixed eleven times.

Headline findings

- **Q-01 / Q-02 — Mobile nav broken:** A CSS `!important` rule in `global.css:277` (`.desktop-nav { display: none !important }`) overrides the inline `style.display = 'flex'` toggle in `Nav.astro:118`. CSS cascade beats inline non-`!important`. Tapping the hamburger does nothing. There are also no mobile drawer styles (full-width, stacked links, visible panel) — even if the toggle worked, the layout would be wrong.
- **Q-03 — FAQ accordion broken:** The JS adds class `.open`, but the CSS only styles `.faq-answer` (without `.open`) with `max-height: 0` and a `max-height` transition. The transition never fires; the panel stays hidden.
- **Q-07 — Two illegal CSS transitions:** `transition: width` on the blog reading progress bar (triggers layout/paint on every scroll event — visibly lags on mobile); `transition: max-height` on the FAQ accordion (the second-class citizen of CSS transitions). Both violate `CLAUDE.md`'s `transform / opacity`-only rule.
- **Q-08 — Desktop-first CSS, despite "mobile-first" declared:** Every `@media` query across all files uses `max-width` (override-down). Zero `min-width` queries (enhance-up). This is the most leveraged finding because it touches every component.
- **Q-12 — WCAG AA contrast fail:** Footer secondary text `#666` on `#1C1C1C` $\approx 3.0:1$. Needs $\geq 4.5:1$. Use `#999` or lighter.

CLAUDE.md drift surfaced by QA audit

Three rules in `CLAUDE.md` are claimed but not enforced:

Rule	Reality	Severity
"Mobile-first responsive"	Entire CSS system is desktop-first (12 <code>max-width</code> queries, 0 <code>min-width</code>)	High
"Only animate <code>transform</code> and <code>opacity</code> "	Two violations: <code>transition: width</code> , <code>transition: max-height</code>	High
"Every clickable element needs hover, focus-visible, and active states. No exceptions."	<code>.dropdown-item</code> and <code>.faq-question</code> are missing <code>:active</code> states	High

These three drift items become Section 4 entries D-11 through D-13.

Standardization proposal (the 30-minute / 11× leverage move)

The QA audit's Section 7 proposes a centralized responsive system added to the top of `src/styles/global.css`: three breakpoints (640px mobile, 1024px tablet, 1280px desktop), three utility classes (`.grid-auto-3`, `.grid-auto-2`, `.container-page`), a typography scale using `clamp()`, mobile-first media queries throughout, and a working mobile nav drawer. It also proposes a `CLAUDE.md` addition under "Output Defaults" so future page authors don't introduce new ad-hoc breakpoints.

Why this matters for sequencing: Task #5 will replicate the AI Trends pattern across 10 pages plus the homepage. Every ad-hoc responsive pattern fixed today is fixed once. After Task #5 lands without centralization, the same fix costs 11×. This is the same "fix-once-vs-fix-eleven-times" argument that justified pulling the SEO `seo` object into Task #0b for the May 7 cycle. The mobile case has identical mechanics.

What QA audit did NOT cover (scope boundary)

The audit was code analysis only — no live browser testing (the site isn't deployed). Lighthouse mobile scores, real-device Safari/Chrome divergence, screen-reader navigation, and keyboard tab order are post-deploy items recorded in the QA audit's Section 4 "Post-Launch Optimization Roadmap" (priorities 1–7).

4. Drift Report (Documented Claims vs. Actual State)

May 7 surfaced ten drift items (D-01 through D-10). The QA audit adds three more — all three are `CLAUDE.md` self-violations. One existing item (D-03) is updated with new evidence.

#	Source	Claim	Reality	Action
D-01	<code>summary.md</code>	"Adding a language requires one constant + Sanity content"	Partially false. The <code>locale</code> ternary in two blog pages also requires a code change	Fix F-05 in #0e
D-02	<code>summary.md</code>	"Three EN posts live in Sanity"	True for EN; <code>/es/blog</code> renders an empty list — not flagged in status table	Either backfill ES content or update status table

#	Source	Claim	Reality	Action
D-03	summary.md	"AI Trends page — content-complete"	Now extended: Three hardcoded English strings appear on /es/ai-trends (technical audit), AND the FAQ accordion does not open at all on any device (Q-03 from QA audit). Calling this page "content-complete" is doubly inaccurate	Fix in #0 / #5
D-04	summary.md	Status table marks Task #8 as "Pending"	Confirmed accurate	None
D-05	summary.md claim "hreflang via i18n routing"	URL structure alone does not emit hreflang tags	Update to "URL structure ready; hreflang tags pending #0c"	
D-06	summary.md	"Language switcher links to equivalent page"	Always routes to /{lang}/	Update to "Switcher routes to homepage; per-page equivalence pending #10"
D-07	go-live-guide.md Phase 6.2	References amplify.yml build spec	amplify.yml does not exist at repo root	Create file in #8
D-08	go-live-guide.md Phase 6.4	Sanity webhook trigger described	Not configured anywhere	None — doc is forward-looking spec
D-09	.gitignore	.env excluded by rule	.env is currently tracked	Run git rm --cached .env in #0a
D-10	(no document flags this)	imageMeta.ts extends the sanity.imageAsset system type	Non-standard; could break on Sanity major upgrades	Document in summary.md schema notes (#11)
D-11	CLAUDE.md	"Mobile-first responsive"	All CSS is desktop-first — 12 @media (max-width) queries, zero @media (min-width) queries	Refactor in #0f using QA audit's Section 7 standardization proposal

#	Source	Claim	Reality	Action
D-12	CLAUDE.md	"Only animate transform and opacity "	Two violations: transition: width on blog reading bar; transition: max-height on FAQ accordion	Replace in #0f: transform: scaleX() for the bar, JS height animation for the FAQ
D-13	CLAUDE.md	"Every clickable element needs hover, focus-visible, and active states. No exceptions."	.dropdown-item and .faq-question are missing :active states	Add in #0f

Thirteen drift items total (was 10). All thirteen are fixable in under 1.5 cumulative days. The CLAUDE.md self-violations (D-11–D-13) are particularly important because the project-rules file is what Claude Code and any human contributor consult before writing frontend code — drift between rules and reality compounds.

5. Recommended Roadmap (Mapped to Existing Task #0–12 Plan)

The May 7 audit produced audits/IMPLEMENTATION_PLAN.md , which is the canonical task tracker (it supersedes the "Next, in order" block at the top of summary.md). This synthesis updates the plan in **one place only**: Task #0 gains a new sub-item #0f.

Task #0 — Pre-flight (NEW since May 7; expands in this synthesis)

The existing five sub-items (#0a–e) hold. The QA/mobile audit adds one consolidated sub-item:

ID	Sub-task	Audit ref	Effort	Blocks
#0a	git rm --cached .env , add .env.example , commit	F-01, D-09	S (5 min)	#8
#0b	Add seo object to aiTrendsPage and post schemas	S-09	M (1–2 hr)	#5
#0c	Add canonical / hreflang / Open Graph / Twitter Card to BaseLayout.astro	S-02/03/05/06/07/08	M (3–4 hr)	#5, #8
#0d	Install @astrojs/sitemap , set site URL, create public/robots.txt	S-01, S-04	M (45 min)	#8

ID	Sub-task	Audit ref	Effort	Blocks
#0e	Fix EN/ES delocalization: <code>locale</code> in <code>LANG_META</code> , dynamic <code>language</code> source for <code>post.ts</code>	F-05, A-07, D-01	M (2–3 hr)	#10
#0f	Visual / Mobile / A11y pre-flight (NEW): centralize responsive breakpoint system in <code>global.css</code> per QA audit §7; fix mobile nav <code>!important</code> override (Q-01) and add drawer styles (Q-02); fix FAQ accordion (Q-03); replace <code>transition: width</code> and <code>transition: max-height</code> with allowed properties (Q-07); add <code>:active</code> states (Q-09); hamburger 44×44 hit area (Q-05); footer contrast (Q-12); <code>prefers-reduced-motion</code> (Q-16); <code>defer</code> on Tailwind CDN script (Q-15); update <code>CLAUDE.md</code> with the Responsive System section per QA audit §7	Q-01–Q-09, Q-12, Q-15, Q-16, Q-17; D-11/12/13	M (4–6 hr)	#5

Also fix at the same time (unchanged from May 7): F-07 null-guard, F-06/S-16 alt-text validation, F-02/F-03/F-04 hardcoded English strings, F-13 default description, S-18 `noindex`.

Total Task #0 effort: 1.5–2.5 days (was 1–2). Do not start Task #5 until this is green.

Tasks #5 through #12 — no other re-sequencing required

The QA findings either fit inside Task #0f (almost all) or land naturally inside Task #9 post-deploy (raw `<img>` → Astro `<Image>` migration; full Tailwind CDN → build-step migration; Lighthouse mobile real-device testing). No new tasks are warranted — the supervisor explicitly reviewed whether to propose a Task #13 and concluded it would create a parallel todo list rather than fitting findings inside the existing structure.

Specifically:

- **Task #5** — Pattern hardening from #0f means each batch automatically inherits a working mobile nav, the centralized breakpoint system, valid CSS transitions, complete interactive states, and proper touch targets. No scope change beyond the existing May 7 expansion.
- **Task #6** — No QA-driven changes. Bulk migration scripts and AI-assisted content remain as scoped.
- **Task #8** — No QA-driven changes. Webhook + `amplify.yml` + env vars + monitoring as scoped May 7.
- **Task #9** — Two QA items move here as the *post-deploy* polish layer: full Astro `<Image>` migration (Q-13 reinforces S-17) and a Lighthouse-mobile baseline run.
- **Task #10** — Unchanged. F-05/A-07 prerequisites still gate language 3.
- **Task #11** (NEW from May 7) — Operations Safety Net. Unchanged.
- **Task #12** (NEW from May 7) — AI Ops Pipeline. Unchanged.

6. New Tasks (none added in this synthesis)

The May 7 audit added Tasks #11 (Operations Safety Net) and #12 (AI Ops Pipeline). The QA/mobile audit added in this synthesis did **not** surface anything that warrants another new task. Every QA finding maps cleanly to Task #0f (pre-flight), Task #5 (pattern replication), or Task #9 (post-deploy polish). This is a deliberate choice — the supervisor's job is to fit findings into the existing 12-task structure where possible, and a new "Task #13: Mobile Stuff" would dilute that discipline.

The QA audit's *post-launch optimization roadmap* (priorities 1–7 in its Section 4) is explicitly post-launch work and naturally lives inside Task #9's tail end.

7. Decision Justification

Recommendation: GO WITH CAVEATS — unchanged from May 7.

The QA/mobile audit reinforces the May 7 conclusion. The trajectory is still credible: 12 sessions of real progress, sound architecture, locked-in tech stack, no findings that justify HOLD or PIVOT. What changed is that a fourth specialist looked specifically for the things the first three didn't cover, and found one launch-blocker cluster (mobile navigation completely broken) plus three `CLAUDE.md` self-violations. None of these are architectural — they are implementation gaps inside the locked-in design choices, which is exactly the category the audit cycle exists to surface.

The QA findings actually *strengthen* the case for the pre-flight pattern that Task #0 institutionalized in the May 7 plan. The May 7 logic was: "Don't replicate the AI Trends singleton pattern across 11 pages until the SEO and EN/ES gaps are closed in the base — otherwise we fix the same problem 11 times." The QA audit shows the same logic applies to the responsive system: a 30-minute centralization in `global.css` today saves eleven copies of the same broken responsive code tomorrow. This is high-leverage work being caught at exactly the right moment in the project lifecycle, which is the value an audit cycle is meant to produce. If the QA audit had run *after* Task #5, the cost would have been roughly 8–10× higher.

The cost picture remains the supporting argument. Today's burn is ~\$1.25/month; full 15-language scale stays under \$60/month. The largest hidden cost — manual content entry without bulk migration scripts — is a 80-to-13-hour delta that dwarfs any infrastructure decision. The QA additions to Task #0 cost approximately 4–6 hours of pre-flight work and prevent eleven copies of broken mobile navigation. Approve continued commitment, conditional on the expanded 30-day priority list landing on schedule. The decision in front of the boss is not "is this worth keeping" — it is "do we close the four specific gaps above (security + SEO foundation + EN/ES delocalization + Visual/Mobile/A11y) before the next push?" The supervisor's answer is yes; the work is bounded; the audit cycle is delivering exactly the early-warning value it was designed to.

Appendix A — Audit Sources

- `audits/technical-audit.md` (2026-05-07) — Routing, Sanity schemas, i18n, Sanity-first compliance, build config, drift check
- `audits/seo-audit.md` (2026-05-07) — Indexability, multilingual structure, metadata, structured data, images, performance, migration
- `audits/automation-audit.md` (2026-05-07) — Build pipeline, deploy triggers, env management, bulk migration, AI ops, cost, monitoring, backup
- `audits/qa-mobile-audit.md` (2026-05-08, NEW) — **Responsive behavior, mobile navigation, touch targets, CLAUDE.md drift, accessibility, interactive states**
- `summary.md` — current project state and session-by-session changelog
- `go-live-guide.md` — phase-by-phase deployment roadmap
- `CLAUDE.md` — project rules and design guardrails
- `audits/IMPLEMENTATION_PLAN.md` — canonical task tracker (post-May-7 audit)
- `audits/FINAL_PROJECT_AUDIT_2026-05-07.md` — archived prior synthesis (for diff reference)

Appendix B — Task Tracker Status After This Synthesis

#	Task	Status	Net Audit-Driven Change (May 7 + 2026-05-08)
1	Restructure repo	Done	No change
2	Dynamic Nav language switcher	Done	No change (S-14 / per-page equivalence stays in #10)
3	Sanity schema POC for AI Trends	Done	No change
4	Implement Sanity POC	Done	No change
0	Pre-flight (NEW from May 7; expanded in 2026-05-08)	Pending	6 sub-items: #0a–#0e from May 7; #0f Visual/Mobile/A11y NEW. 1.5–2.5 days.
5	Port remaining 11 pages (Batch A → B → C)	Blocked by #0	Pattern hardening from #0 includes mobile/A11y from this synthesis
6	Enter EN + ES content + bulk migration scripts	Parallel to #5	No change from May 7
7	Update <code>go-live-guide.md</code> for AWS Amplify	Done	No change

#	Task	Status	Net Audit-Driven Change (May 7 + 2026-05-08)
8	Deploy to AWS Amplify with Sanity webhook	Pending; gated by #0a	No change from May 7
9	SEO + performance polish (post-deploy)	Pending	Adds: Q-13 image migration alongside S-17; Q-15 full Tailwind CDN → build-step migration; Lighthouse mobile baseline
10	Scale to languages 3–15	Pending; gated by #0e	No change from May 7
11	Operations Safety Net (NEW from May 7)	Proposed	No change
12	AI Ops Pipeline (NEW from May 7)	Proposed	No change

End of report. This synthesis is dated 2026-05-08. The prior May 7 version is archived at [audits/FINAL_PROJECT_AUDIT_2026-05-07.md](#) for diff comparison.